

Consumer Preference and Buying Behavior Toward Packaged Tea

Brands

in

Bangladesh

This project was conducted to analyze consumer preference, purchasing behavior, satisfaction, and brand loyalty toward packaged tea brands in Bangladesh using statistical analysis techniques in SPSS. The study focused on understanding how consumers choose tea brands and how factors such as taste, packaging, advertisement, and income influence buying behavior in the FMCG sector.

A structured dataset consisting of 250 respondents was used for the analysis. The dataset was prepared for analytical and academic purposes based on realistic consumer purchasing patterns. Variables included demographic information (gender, age, occupation, income) and consumer behavior variables such as preferred tea brand, purchase frequency, advertisement influence, taste satisfaction, packaging satisfaction, brand loyalty, and recommendation intention.

Several statistical techniques were applied in the project, including Frequency Analysis, Descriptive Statistics, Crosstab Analysis, Chi-Square Test, Correlation Analysis, Independent Sample T-Test, One-Way ANOVA, and Reliability Analysis (Cronbach's Alpha). The findings revealed that Ispahani was the most preferred tea brand among respondents, while taste and packaging satisfaction showed strong positive relationships with brand loyalty.

This project demonstrates how statistical analysis can be used to generate meaningful consumer insights and support market research and strategic decision-making in the FMCG industry.

Data View

Visible: 14 of 14 Variable											
ID	Gender	Age_Group	Occupation	Monthly_Income	Preferred_Brand	Purchase_Frequency	Buying_Place	Advertisement_Influence	Price_Sensitivity	Taste_Satisfaction	Packaging_Satisfaction
19	Female	18-25 Years	Student	20,000-40,000 BDT	Tetley	Weekly	Local Store	Strongly Agree	Agree	Disagree	Disagree
20	Male	18-25 Years	Business	40,000-60,000 BDT	Tetley	Monthly	Local Store	Agree	Strongly Agree	Agree	Strongly Agree
21	Female	18-25 Years	Student	Below 20,000 BDT	Fresh	Daily	Local Store	Strongly Disagree	Agree	Agree	Strongly Agree
22	Female	18-25 Years	Service Holder	Below 20,000 BDT	Ispahani	Weekly	Local Store	Agree	Neutral	Agree	Agree
23	Female	18-25 Years	Service Holder	Below 20,000 BDT	Ispahani	Monthly	Local Store	Neutral	Agree	Agree	Neutral
24	Female	18-25 Years	Homemaker	40,000-60,000 BDT	Fresh	Weekly	Local Store	Neutral	Disagree	Neutral	Disagree
25	Male	26-35 Years	Student	20,000-40,000 BDT	Ispahani	Weekly	Local Store	Agree	Neutral	Disagree	Neutral
26	Male	26-35 Years	Student	20,000-40,000 BDT	Tetley	Weekly	Local Store	Agree	Agree	Agree	Strongly Agree
27	Female	36-45 Years	Student	Below 20,000 BDT	Ispahani	Daily	Super Shop	Strongly Agree	Agree	Agree	Agree
28	Female	36-45 Years	Service Holder	40,000-60,000 BDT	Ispahani	Daily	Local Store	Strongly Agree	Strongly Agree	Strongly Agree	Strongly Agree
29	Male	46+ Years	Business	20,000-40,000 BDT	Finlay	Weekly	Super Shop	Neutral	Strongly Disagree	Strongly Agree	Strongly Agree
30	Female	18-25 Years	Student	Above 60,000 BDT	Others	Weekly	Local Store	Neutral	Strongly Agree	Agree	Strongly Agree
31	Female	36-45 Years	Student	20,000-40,000 BDT	Ispahani	Weekly	Local Store	Disagree	Disagree	Strongly Agree	Agree
32	Male	26-35 Years	Student	40,000-60,000 BDT	Finlay	Daily	Super Shop	Neutral	Strongly Agree	Strongly Agree	Strongly Agree
33	Female	36-45 Years	Student	Below 20,000 BDT	Finlay	Weekly	Local Store	Strongly Disagree	Disagree	Strongly Agree	Strongly Agree
34	Male	26-35 Years	Service Holder	20,000-40,000 BDT	Others	Daily	Super Shop	Neutral	Strongly Agree	Strongly Agree	Strongly Agree
35	Male	26-35 Years	Homemaker	20,000-40,000 BDT	Fresh	Weekly	Local Store	Strongly Disagree	Agree	Agree	Strongly Agree
36	Male	18-25 Years	Service Holder	20,000-40,000 BDT	Fresh	Daily	Online	Agree	Strongly Agree	Strongly Agree	Strongly Agree
37	Female	18-25 Years	Homemaker	40,000-60,000 BDT	Tetley	Weekly	Local Store	Strongly Agree	Agree	Agree	Strongly Agree
38	Female	36-45 Years	Business	20,000-40,000 BDT	Ispahani	Daily	Super Shop	Strongly Agree	Strongly Agree	Strongly Agree	Agree
39	Female	18-25 Years	Service Holder	Below 20,000 BDT	Others	Weekly	Super Shop	Disagree	Strongly Agree	Agree	Neutral
40	Male	36-45 Years	Service Holder	20,000-40,000 BDT	Tetley	Daily	Super Shop	Agree	Neutral	Disagree	Agree
41	Male	18-25 Years	Student	20,000-40,000 BDT	Ispahani	Daily	Online	Neutral	Strongly Agree	Agree	Strongly Agree
42	Male	36-45 Years	Service Holder	Below 20,000 BDT	Ispahani	Monthly	Local Store	Agree	Disagree	Agree	Agree
43	Male	18-25 Years	Student	20,000-40,000 BDT	Ispahani	Daily	Local Store	Neutral	Disagree	Agree	Neutral
44	Male	26-35 Years	Student	20,000-40,000 BDT	Ispahani	Weekly	Super Shop	Agree	Agree	Strongly Agree	Strongly Agree

Variable View

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Descriptive and Frequency Analysis:

Statistics								
		Gender of Respondent	Age Category of Respondent	Occupation Status	Monthly Income Category	Preferred Tea Brand	Frequency of Tea Purchase	Main Place of Purchase
N	Valid	250	250	250	250	250	250	250
	Missing	0	0	0	0	0	0	0

Interpretation: A total of 250 valid responses were included in the analysis for all variables, with no missing data found in the dataset. This indicates that the dataset was complete and suitable for further statistical analysis.

Gender of Respondent					
		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Male	142	56.8	56.8	56.8
	Female	108	43.2	43.2	100.0
	Total	250	100.0	100.0	

Among the 250 respondents, 56.8% were male and 43.2% were female. This indicates that male respondents slightly dominated the study sample.

Age Category of Respondent					
		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	18-25 Years	103	41.2	41.2	41.2
	26-35 Years	88	35.2	35.2	76.4
	36-45 Years	41	16.4	16.4	92.8
	46+ Years	18	7.2	7.2	100.0

	Total	250	100.0	100.0	
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The majority of respondents (41.2%) belonged to the 18–25 years age group, followed by 35.2% from the 26–35 years category. This indicates that young consumers represented the largest portion of the study sample.

Occupation Status					
		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Student	96	38.4	38.4	38.4
	Service Holder	91	36.4	36.4	74.8
	Business	34	13.6	13.6	88.4
	Homemaker	29	11.6	11.6	100.0
	Total	250	100.0	100.0	

Most respondents were students (38.4%) and service holders (36.4%), while smaller proportions belonged to business and homemaker categories. This indicates that the study mainly represented young and working consumers.

Monthly Income Category					
		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Below 20,000 BDT	67	26.8	26.8	26.8
	20,000-40,000 BDT	105	42.0	42.0	68.8
	40,000-60,000 BDT	56	22.4	22.4	91.2
	Above 60,000 BDT	22	8.8	8.8	100.0
	Total	250	100.0	100.0	

The largest proportion of respondents (42.0%) had a monthly income between 20,000–40,000 BDT, followed by 26.8% earning below 20,000 BDT. This suggests that most respondents belonged to the middle-income group.

Preferred Tea Brand					
		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Ispahani	113	45.2	45.2	45.2
	Fresh	54	21.6	21.6	66.8
	Finlay	38	15.2	15.2	82.0
	Tetley	31	12.4	12.4	94.4
	Others	14	5.6	5.6	100.0
	Total	250	100.0	100.0	

Ispahani was the most preferred tea brand among respondents (45.2%), followed by Fresh (21.6%) and Finlay (15.2%). This indicates that Ispahani held the strongest consumer preference in the study.

Frequency of Tea Purchase					
		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Daily	103	41.2	41.2	41.2
	Weekly	102	40.8	40.8	82.0
	Monthly	45	18.0	18.0	100.0
	Total	250	100.0	100.0	

Most respondents purchased tea either daily (41.2%) or weekly (40.8%), while only 18.0% purchased tea monthly. This indicates that tea is a frequently consumed FMCG product among consumers.

Main Place of Purchase					
		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Super Shop	90	36.0	36.0	36.0
	Local Store	140	56.0	56.0	92.0
	Online	20	8.0	8.0	100.0
	Total	250	100.0	100.0	

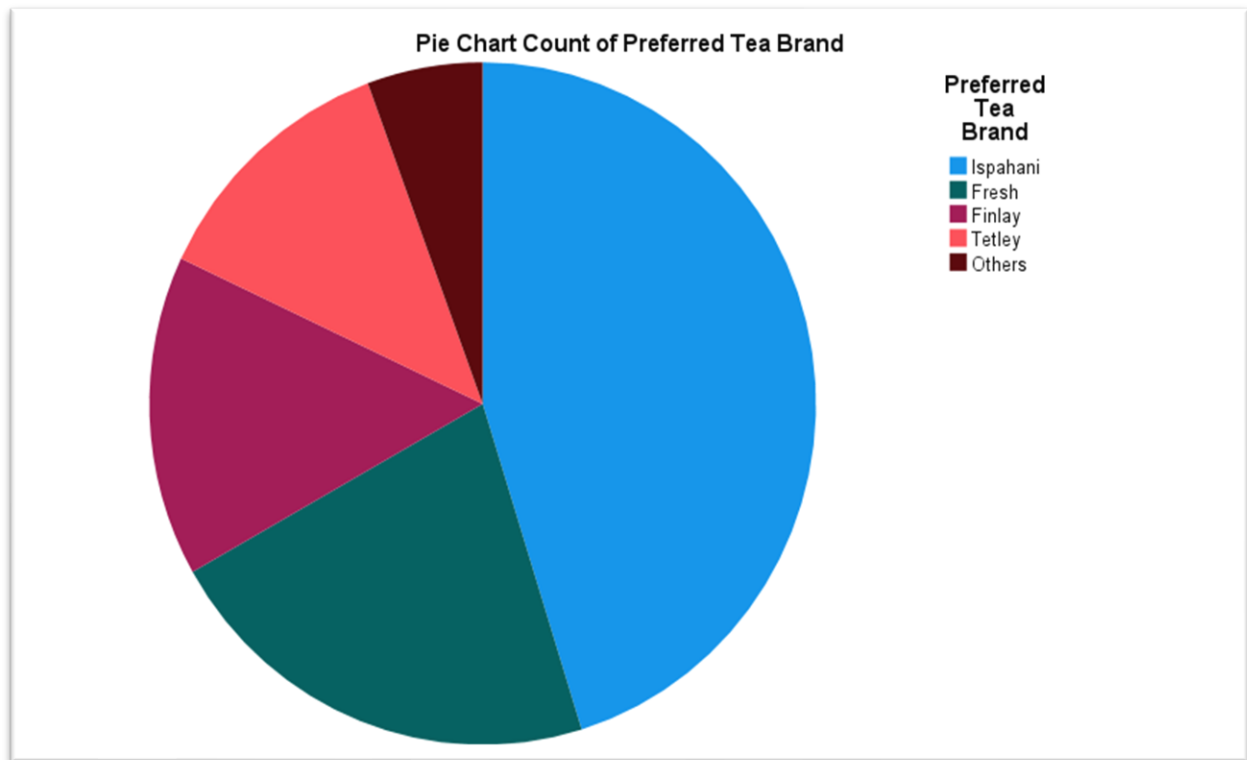
The majority of respondents (56.0%) purchased tea from local stores, followed by 36.0% from super shops, while only 8.0% purchased online. This suggests that local stores remain the primary purchasing channel for tea products.

Descriptive Statistics					
	N	Minimum	Maximum	Mean	Std. Deviation
Influence of Advertisement	250	1	5	3.60	1.018
Sensitivity Toward Product Price	250	1	5	3.77	1.058
Satisfaction with Taste	250	2	5	3.81	.898
Satisfaction with Packaging	250	1	5	3.74	1.098
Brand Loyalty Level	250	1	5	3.76	1.021
Intention to Recommend Brand	250	1	5	3.67	.963
Valid N (listwise)	250				

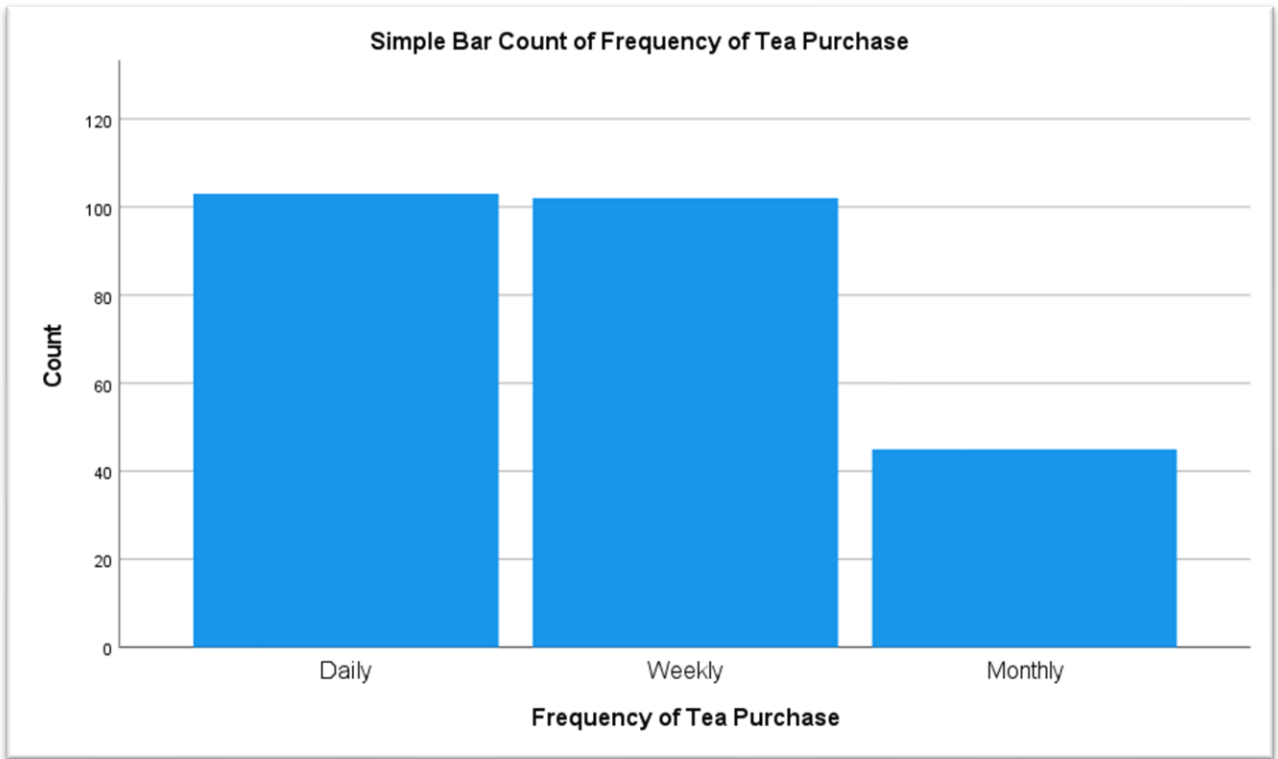
The descriptive statistics indicate that respondents showed moderate to high agreement regarding advertisement influence, price sensitivity, taste satisfaction, packaging satisfaction, brand loyalty, and recommendation intention. Among all variables, taste satisfaction had the highest mean score (Mean = 3.81), suggesting that consumers were generally satisfied with the

taste of their preferred tea brands. The standard deviation values indicate moderate variation in respondents' opinions across all variables.

Charts:



The pie chart illustrates the distribution of preferred tea brands among respondents. Ispahani occupied the largest portion of the chart, indicating that it was the most preferred tea brand among consumers, followed by Fresh and Finlay. This suggests that Ispahani holds a strong market preference compared to competing brands.



The bar chart shows that most respondents purchased tea either daily or weekly, while comparatively fewer respondents purchased tea monthly. This indicates that tea is a regularly consumed FMCG product among consumers.

Gender of Respondent * Preferred Tea Brand Crosstabulation								
			Preferred Tea Brand					Total
			Ispahani	Fresh	Finlay	Tetley	Others	
Gender of Respondent	Male	Count	68	28	23	15	8	142
		% within Gender of Respondent	47.9%	19.7%	16.2%	10.6%	5.6%	100.0%
		% within Preferred Tea Brand	60.2%	51.9%	60.5%	48.4%	57.1%	56.8%

	Female	Count	45	26	15	16	6	108
		% within Gender of Respondent	41.7%	24.1%	13.9%	14.8%	5.6%	100.0%
		% within Preferred Tea Brand	39.8%	48.1%	39.5%	51.6%	42.9%	43.2%
Total		Count	113	54	38	31	14	250
		% within Gender of Respondent	45.2%	21.6%	15.2%	12.4%	5.6%	100.0%
		% within Preferred Tea Brand	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%

The Crosstab analysis shows that Ispahani was the most preferred tea brand among both male (47.9%) and female (41.7%) respondents. Although slight differences in brand preference were observed between genders, both groups showed a strong preference for Ispahani compared to other brands.

Chi-Square Tests			
	Value	df	Asymptotic Significance (2-sided)
Pearson Chi-Square	2.174 ^a	4	.704
Likelihood Ratio	2.166	4	.705
Linear-by-Linear Association	.570	1	.450
N of Valid Cases	250		
a. 0 cells (.0%) have expected count less than 5. The minimum expected count is 6.05.			

The Chi-Square test result showed a p-value of 0.704, which is greater than the significance level of 0.05. Therefore, there was no statistically significant association between gender and preferred tea brand among the respondents.

Correlation Analysis:

Correlations						
		Satisfaction with Taste	Satisfaction with Packaging	Brand Loyalty Level	Intention to Recommend Brand	Influence of Advertisement
Satisfaction with Taste	Pearson Correlation	1	.731**	.707**	.280**	-.032
	Sig. (2-tailed)		.000	.000	.000	.609
	N	250	250	250	250	250
Satisfaction with Packaging	Pearson Correlation	.731**	1	.739**	.397**	.046
	Sig. (2-tailed)	.000		.000	.000	.471
	N	250	250	250	250	250
Brand Loyalty Level	Pearson Correlation	.707**	.739**	1	.463**	.015
	Sig. (2-tailed)	.000	.000		.000	.819
	N	250	250	250	250	250
Intention to Recommend Brand	Pearson Correlation	.280**	.397**	.463**	1	.450**
	Sig. (2-tailed)	.000	.000	.000		.000
	N	250	250	250	250	250
Influence of Advertisement	Pearson Correlation	-.032	.046	.015	.450**	1
	Sig. (2-tailed)	.609	.471	.819	.000	
	N	250	250	250	250	250

** . Correlation is significant at the 0.01 level (2-tailed).

The correlation analysis revealed strong positive relationships between taste satisfaction, packaging satisfaction, and brand loyalty. In particular, packaging satisfaction showed the strongest positive correlation with brand loyalty ($r = 0.739$, $p < 0.01$), indicating that consumers who were more satisfied with packaging and taste were more likely to remain loyal to the brand. Additionally, advertisement influence had a moderate positive relationship with recommendation intention ($r = 0.450$, $p < 0.01$).

Independent Samples Test:

Independent Samples Test										
		Levene's Test for Equality of Variances		t-test for Equality of Means						
		F	Sig.	t	df	Sig. (2-tailed)	Mean Difference	Std. Error Difference	95% Confidence Interval of the Difference	
									Lower	Upper
Satisfaction with Taste	Equal variances assumed	.117	.733	-.815	248	.416	-.094	.115	-.319	.132
	Equal variances not assumed			-.814	229.859	.416	-.094	.115	-.320	.133

The Independent Sample T-Test was conducted to compare taste satisfaction between male and female respondents. Levene's Test showed a significance value of 0.733, which is greater than 0.05, indicating that the assumption of equal variances was satisfied. The t-test result revealed a

p-value of 0.416, which is greater than 0.05, suggesting that there was no statistically significant difference in taste satisfaction between male and female respondents.

ANOVA:

ANOVA					
Brand Loyalty Level					
	Sum of Squares	df	Mean Square	F	Sig.
Between Groups	3.962	3	1.321	1.271	.285
Within Groups	255.638	246	1.039		
Total	259.600	249			

The One-Way ANOVA test was conducted to examine whether brand loyalty differed across different income groups. The result showed a p-value of 0.285, which is greater than 0.05, indicating that there was no statistically significant difference in brand loyalty among the income groups of respondents.

Multiple Comparisons						
Dependent Variable: Brand Loyalty Level						
Tukey HSD						
(I) Monthly Income Category	(J) Monthly Income Category	Mean Difference (I-J)	Std. Error	Sig.	95% Confidence Interval	
					Lower Bound	Upper Bound
Below 20,000 BDT	20,000-40,000 BDT	-.115	.159	.889	-.53	.30

	40,000-60,000 BDT	-.058	.185	.989	-.53	.42
	Above 60,000 BDT	-.480	.250	.224	-1.13	.17
20,000-40,000 BDT	Below 20,000 BDT	.115	.159	.889	-.30	.53
	40,000-60,000 BDT	.057	.169	.987	-.38	.49
	Above 60,000 BDT	-.365	.239	.423	-.98	.25
40,000-60,000 BDT	Below 20,000 BDT	.058	.185	.989	-.42	.53
	20,000-40,000 BDT	-.057	.169	.987	-.49	.38
	Above 60,000 BDT	-.422	.256	.355	-1.09	.24
Above 60,000 BDT	Below 20,000 BDT	.480	.250	.224	-.17	1.13
	20,000-40,000 BDT	.365	.239	.423	-.25	.98
	40,000-60,000 BDT	.422	.256	.355	-.24	1.09

The Tukey HSD post hoc test showed that all pairwise comparisons between income groups had p-values greater than 0.05. This indicates that there was no statistically significant difference in brand loyalty between any two income groups.